

THE 75 DOLLAR BOX FAN SMOKE CLEANER: EVERY NUMBER

A 20 inch box fan and four furnace filters taped into a cube clear wildfire smoke about as well as a 650 dollar purifier, at roughly a tenth of the sticker price. Here is every figure from the video, hedges kept.

WHAT IT IS

Stand four pleated furnace filters on edge to make a square, cap it with a box fan on top, add a cardboard floor and a cardboard top shroud. The fan pulls room air in through all four filter walls and blows clean air out the top. Clean air delivery, measured in cubic feet per minute (cfm), is how much smoke free air the machine makes each minute. Bigger is better.

The Environmental Protection Agency tested this in a smoke chamber, using smoke from smoldering pine needles.

THE COMPARISON IN NUMBERS

401 cfm	Clean air the four filter cube delivered in the agency chamber. It carries a margin of about 31 either way.
111 cfm	A single filter on the same fan, no shroud, in the same tests.
649 dollars	Coway lists the Airmega 400 purifier at this price, with a smoke rating of 328. That is the maker's own number.
about 75	Fair cost to plan around for the box build.

Different labs tested the two machines with different smoke, so call it a tie at about a tenth of the price. These are sticker prices; replacement filters change the math over a season.

COST PER UNIT OF CLEAN AIR

19 cents	Roughly what the box costs for each cubic foot per minute of clean air. Price divided by airflow.
2 dollars	Roughly what the purifier costs for the same thing. About ten times more.

WHY FILTERING SMOKE MATTERS

2.5 microns	Wildfire smoke carries particles this size or smaller. They reach deep into your lungs, and some may get into your bloodstream.
70 microns	Roughly the width of a human hair, so the biggest smoke particles are around thirty times thinner.
1 day	The agency says a single day of exposure can do harm. Higher risk: children, older adults, pregnant women, and people with heart or lung disease.
24,000	Deaths a year across the contiguous United States that a Science Advances study, covering 2006 through 2020, tied to wildfire smoke particles. It is a statistical model of long term exposure, not a count of people caught in any one fire.
Merv 13	Merv is a filter grade. The agency recommends grade 13 for smoke. It catches at least half of the smallest smoke particles on every pass, and most larger ones.

Closed windows slow smoke down but do not stop it. A good filter catches those particles.

Scoreboard	Box vs purifier, in the lab
THE BOX 401 cfm in the chamber for about 75 dollars. Roughly 19 cents per unit of clean air.	THE PURIFIER Rated 328 for 649 dollars. Roughly 2 dollars per unit, about ten times the cost. It does arrive with a measured rating and needs no build.

THE PARTS AND WHAT THEY COST

20 to 30	Dollars for the 20 inch box fan. Lasko lists its basic 20 inch fan at about 30 dollars.
35 to 55	Dollars for four pleated furnace filters, 20 by 20 by 1, rated Merv 13 or higher.
5 to 8	Dollars for duct tape.
free	Cardboard comes from the fan's own shipping box.
60 to 93	The low ends add up to 60 dollars, the high ends to 93. About 75 is a fair number to plan around.

Prices vary by store and brand; the fan range comes from one filter seller's parts list. Cheaper filter grades sit on the same shelf, so read the label before you buy. They are not built to catch the smallest smoke particles.

FAN AGE GATE

The fan must be built in 2012 or later, with a safety mark from a testing lab printed on its label. The agency says older fans may pose a fire risk. If yours is older, buy a new one for 20 to 30 dollars and leave the old one out of this.

BUILDING IT, STEP BY STEP

The build takes about 20 minutes. Check your filter size first: a filter labeled 20 by 20 by 1 is not really 20 inches. One brand measures about nineteen and a half inches square, so the fan overhangs a little and tape closes the gap.

- 1 Stand the four filters on edge to make a square, with the airflow arrow on each frame pointing inward.
- 2 Tape every corner seam, inside and out, along its whole length.
- 3 Cut a cardboard floor to fit under the cube, and tape it down on all four sides.
- 4 Set the fan on top, facing up. Tape its frame to the top edges of the filters all the way around.
- 5 Make the shroud: cut a cardboard square that covers the top face of the fan, cut a circle in the middle the size of the fan blades, and tape it on. Air can only leave through the center.
- 6 Check for leaks. Hold a tissue near the seams while the fan runs, and add tape wherever it gets pulled toward a gap.

Clamps and bungee cords are other ways to hold the cube together, but tape does the sealing. Any air that sneaks past the filters goes through untouched.

111 to 156

In the agency's tests, the shroud lifted a single filter from 111 to 156 cfm, about 40 percent more clean air. Leave it off and you land well short of 401.

WHY FOUR FILTERS BEAT ONE

Clean air delivery = airflow through the filter x fraction of particles caught each pass

A filter that catches half the particles, with 400 cfm going through it, delivers 200 of clean air. The real fraction depends on particle size, so treat that as an illustration. A box fan is not built to push through much resistance, so one dense filter in front of it chokes the airflow.

263 cfm	Two filters with the shroud, in the agency's tests.
248 cfm	One thick four inch filter with the shroud.
401 cfm	The four filter cube. About two and a half times the clean air of the single filter with its shroud at 156, from four times the filter.

2.5 sq ft	Surface area of one filter face.
10.5 sq ft	Four filters give the fan this much. A big, easy wall of filter moves a lot of air; lowering the resistance across the surface raised the delivery rate.
960 cu ft	Air in a bedroom 10 feet by 12 with an 8 foot ceiling. At 401 the box could run that whole volume through its filters about every two and a half minutes, in the lab.

WHERE THE BOX FALLS SHORT

The agency calls these a fallback: when commercial air cleaners are not available or affordable, a do it yourself cleaner may offer some protection. Commercial cleaners with a measured rating are typically preferred.

about 10%	Drop in indoor fine particles a small field study in Hoopa, California tied to a box fan with one Merv 13 filter, similar to a small commercial cleaner. It covered eight homes in wildfire smoke and eleven with wood stove smoke. That was one filter, not the cube, so it does not settle what the cube does at home.
52 to 55	Decibels from nine inches for box fan builds on their lowest speed, in a 2022 lab paper.
22 to 52	Decibels Coway lists for the Airmega 400 across its settings. So the box at its quietest is about as loud as the purifier at its loudest.

THE SEASON COST

70 to 110	Dollars for two more filter sets if smoke lingers six weeks and you swap every two weeks, on top of the build.
145 to 185	Dollars in all for that season. The 75 dollar sticker covers only the first set.
6 to 12 mo	How often Coway rates the purifier's own filters. Still well under 649.

Filters load up fast. The agency found do it yourself cleaners almost completely ineffective with dirty filters. Particle filters also cannot remove gases, and smoke carries gases too.

SAFETY RULES FOR THE BUILD

The agency and Underwriters Laboratories ran a fire test in 2021. None of the fan and filter scenarios caught fire, and temperatures met the standard for electric fans. It covered only several box fan models, so these rules still matter.

- ☐ Check the fan label for the safety mark before you tape anything.
- ☐ Plug the fan directly into a wall outlet. No extension cords.
- ☐ Do not use a fan with a damaged cord or housing.
- ☐ Do not leave children unattended while the fan is running.
- ☐ Turn the box off when nobody is home. With a newer certified fan, running it overnight while people are home is fine.
- ☐ Keep it on a stable surface, clear of curtains and bedding.
- ☐ Make sure working smoke detectors are installed throughout your home.

CHANGE THE FILTER IN TIME

Check daily during a smoke event. Change the filter when it turns dark brown or gray, or smells like smoke. In severe smoke that can mean every few weeks or even every few days. Handle used filters gently so the trapped dust stays put, and bag them before you carry them out.

WHO SHOULD SKIP THE BUILD

- If officials order an evacuation, leave. The box is for staying put.
- You need a wall outlet within reach of the fan's own cord; extension cords are off the table.
- A large open room is too much for one box.
- Avoid any purifier that makes ozone on purpose. The agency says no federal agency has approved those for occupied spaces.

SETTING IT UP IN A ROOM

- ☐ Choose the room where you spend the most time, or where someone sleeps.
- ☐ Close the windows and the door.
- ☐ Give the box space away from walls and furniture, so all four sides can pull air.
- ☐ Run it on high while smoke is heavy.
- ☐ Write the install date on each filter frame with a marker.
- ☐ Buy a spare filter set before smoke arrives.

about 88%

Share of particles removed after three passes, if each pass removes half and the air mixes well. A rough picture, and why you leave the box running and the windows shut.

120 - 120

The Association of Home Appliance Manufacturers recommends a smoke rating equal to the room's floor area in square feet. A 120 square foot bedroom needs about 120.

about 400

Square feet the box covers on paper. Real homes did worse than the chamber, so one bedroom is the safe fit.

about 600

Square feet two boxes cover on paper, roughly double the lab rating. A big open living area is where a second box or a larger purifier makes sense.

If your home has central air, the agency also recommends Merv 13 or higher in the furnace or air handler filter slot. Check your system's manual for the highest rating it can handle first. When you go outside in smoke, wear a fitted N95 respirator. An indoor particle monitor will show readings fall once the box is running.

Watch the full video: youtube.com/@SketchySurvival101

More free sheets: sketchysurvival101.com/free

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Sketchy Survival 101. Physics first survival. Every number on this sheet comes straight from the video, with its hedges kept: lab chamber results, maker's own ratings, dated and modeled estimates, and typical values that are not safety limits. A do it yourself cleaner is a fallback, not a replacement for a rated commercial cleaner. Prices move, so check current listings before you act.